Decomposing images in a PRISM: Single-Source Multi-View Framework for Image Representation

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Appendix A. Encoding Maps detail

Appendix A.1. Image to Image

The Fourier transform provides a homogeneous image-to-image mapping that re-expresses the spatial content of an image in the frequency domain [6]. Given an image $I(x, y)$, its two-dimensional discrete Fourier transform is

$$F(u, v) = \sum_{x=0}^{H-1} \sum_{y=0}^{W-1} I(x, y) \exp\left[-2\pi i \left(\frac{ux}{H} + \frac{vy}{W}\right)\right].$$

The resulting representation preserves the regular grid structure of the input while replacing spatial coordinates by frequency components. Low frequencies capture coarse and slowly varying image structure, whereas high frequencies emphasize fine details, edges, and texture. In PRISM, the Fourier representation is used as an alternative view of the same visual signal, allowing hand-designed descriptors to characterize frequency-domain regularities that may be less explicit in the original spatial representation.

Gabor filters provide a localized representation of spatial-frequency content by modulating a sinusoidal carrier with a Gaussian envelope [3]. A bank of filters with different orientations and spatial frequencies can therefore emphasize complementary texture patterns present in the original image. Given an image I, each filter produces

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a response map that preserves the spatial organization of the input while selectively enhancing structures matching a particular orientation and frequency. In PRISM, these response maps are treated as homogeneous image-to-image transformations and are subsequently summarized through simple statistics extracted from the filter responses.

Appendix A.2. Image to Time Series

There are two intuitive ways to convert an image to a time series $I(h, w) \xrightarrow{\sigma_\epsilon} F(t)$. The first treats pixel ordering as an instance of the Traveling Salesman Problem (TSP): pixels are modeled as nodes in a complete graph weighted by spatial distance, and a tour is computed via heuristic search (exact TSP being NP-hard) to approximately minimize total path length while visiting every pixel exactly once.

The resulting ordering's locality preservation depends entirely on the heuristic's quality and carries no closed-form guarantee. The second approach uses a Hilbert curve, a deterministic, recursively constructed space-filling curve that traverses every cell of a grid exactly once, provably guaranteeing that points close in traversal order remain close in the original two-dimensional space [8].

Appendix A.3. Image to Graph

To obtain region-based graph representations, PRISM first partitions the input image into perceptually coherent regions using two complementary superpixel/segmentation strategies [7]: SLIC and the graph-based method of Felzenszwalb and Huttenlocher.

SLIC (Simple Linear Iterative Clustering) [1] groups nearby pixels into compact superpixels by performing a localized clustering in a joint color–spatial space. The method approximately controls the number and compactness of the resulting regions, yielding regular,s spatially localized segments. In our pipeline, each superpixel is subsequently represented as a graph node, while edges encode adjacency between neighboring regions. SLIC therefore induces a relatively homogeneous regional decomposition, emphasizing local spatial organization and appearance consistency.

The Felzenszwalb method follows a different principle [2]. Rather than enforcing approximately regular superpixels, it constructs a graph over pixels and progressively merges components according to the difference between their internal variation and

the weight of edges connecting distinct regions. As a consequence, the resulting segments can vary substantially in size and shape, adapting more directly to local image boundaries and intensity/color discontinuities. When converted into a region-adjacency graph, this segmentation yields a structurally different relational view from SLIC.

Using both segmentations allows PRISM to probe the same image through two distinct inductive biases: SLIC favors compact and approximately uniform regions, whereas Felzenszwalb favors adaptive regions determined by local evidence. Their corresponding graphs therefore provide alternative relational organizations of the same visual signal.

Appendix A.4. Global Maps

Unlike instance-level transformations, which independently map each image to a derived representation, a global map jointly considers multiple instances to construct a shared representation. A natural global representation for an image collection is a similarity graph. Instead of segmenting a single image into regions, each training image becomes a node, and edges encode similarity between different instances. Such a representation makes inter-image regularities explicit, allowing the structure of the dataset itself to become part of the representation.

Let

$$\mathcal{D}_{\text{tr}} = \{(I_i, y_i)\}_{i=1}^{N}$$

denote the training set, and let $\mathbf{z}_i$ be the feature representation associated with image I_i. We construct a k-nearest neighbor graph

$$G_{\text{sim}} = (V, E, W),$$

where each $v_i \in V$ corresponds to one training image and an edge $(v_i, v_j) \in E$ is created whenever I_j belongs to the k nearest neighbors of I_i according to the adopted similarity measure. The resulting affinity matrix W is normalized to obtain a transition matrix

$$P = D^{-1}W, \qquad D_{ii} = \sum_j W_{ij}.$$

Rather than characterizing each node only through its immediate neighborhood, PRISM diffuses the class information over the training graph. Let

$$Y \in \mathbb{R}^{N \times C}$$

be the initial label matrix, where $Y_{ic} = 1$ when training image i belongs to class c and zero otherwise. Starting from $F^{(0)} = Y$, the class distributions are iteratively propagated according to

$$F^{(t+1)} = \alpha P F^{(t)} + (1 - \alpha) Y, \qquad 0 < \alpha < 1.$$

The second term acts as a restart mechanism, retaining a fraction of the initial class evidence at every iteration. At convergence,

$$F^{*} = (1 - \alpha)(I - \alpha P)^{-1} Y,$$

so that each training node is represented by a soft distribution over classes that incorporates evidence propagated through paths of arbitrary length in the graph. The contribution of increasingly distant paths decays geometrically with their length, making the local neighborhood dominant without completely discarding information from the remainder of the connected component.

This construction differs from a conventional k-nearest-neighbor vote. While a kNN classifier considers only the labels of the immediate neighbors of a query, the propagated distribution associated with each training node already summarizes a broader portion of the similarity structure. Thus, a local neighborhood provides access to information that has previously been diffused through the training graph.

Importantly, test images are never inserted into the graph. Given a query image I_q, its feature representation $\mathbf{z}_q$ is used only to retrieve its k nearest training nodes. Its global representation is then obtained by aggregating their converged distributions, for example,

$$\mathbf{g}_q = \frac{1}{k} \sum_{j \in \mathcal{N}_k(q)} F^{*}_j.$$

Consequently, inference does not require rebuilding or re-diffusing the graph when a new image is observed. The construction is therefore inductive with respect to unseen instances, while avoiding test-set contamination because both the graph and the propagated class field are constructed exclusively from training data. No model parameters are optimized during either graph construction or diffusion.

Appendix A.5. Image to Text

The textual view is obtained from a caption generator trained exclusively on natural images, without exposure to artworks or style labels, yet able to express affective tone. Target captions are produced by a large vision-language model T, which receives each training image x together with a fixed prompt p requesting a description of the visible content and of the emotional tone of the scene, with no mention of style, authorship or period:

$$c = T(x, p). \tag{A.1}$$

The resulting set $C = \{(x_i, c_i)\}_{i=1}^{N}$ constitutes the distillation target.

The student model is a decoder-only Transformer that generates the caption autoregressively, conditioned on the embedding of a frozen pretrained visual encoder, $z = \phi(x)$, projected into a set of visual tokens $E(x)$ acting as a prefix:

$$P_\theta(y \mid x) = \prod_{t=1}^{L} P_\theta(y^t \mid y^{<t}, E(x)), \tag{A.2}$$

where $y = \tau(c)$ is the teacher caption after tokenization. The student is trained only on C and therefore never observes artistic images at any point during training.

The parameters θ, comprising the visual projection and the decoder, are fitted by cross-entropy between the student distribution and the teacher caption:

$$\mathcal{L}(\theta) = -\sum_{i=1}^{N} \sum_{t=1}^{L_i} \log P_\theta(y_i^t \mid y_i^{<t}, E(x_i)). \tag{A.3}$$

The objective maximizes the likelihood of the teacher caption under the student model, token by token, so that the student learns to reproduce the conditional distribution of the teacher over descriptions rather than to solve any classification task. Since no style label participates in the optimization and the distillation corpus consists only of

natural images, the textual channel remains independent of the supervision of the target problem. Generation employs decoding with n-gram repetition penalty.

Appendix A.6. Composed maps

Due to Eq. **??**, increasing the transformation depth introduces additional opportunities for information loss, since each successive map operates only on the representation produced by the previous one. However, this does not necessarily imply a corresponding decrease in predictive utility. Under the notion of $\mathcal{V}$-usable information [10], a composed transformation may expose task-relevant structures that were difficult for a constrained classifier to exploit in the preceding representation. To examine this effect, we include two depth-two encoding maps in the evaluation: Horizontal Visibility Graphs (HVGs) and line graphs.

Horizontal Visibility Graphs [4] transform an ordered one-dimensional series into a graph by representing each observation as a node and connecting two observations whenever they have horizontal visibility. More precisely, observations x_i and x_j, with $i < j$, are connected if

$$x_k < \min(x_i, x_j), \qquad \forall\, k \text{ such that } i < k < j. \tag{A.4}$$

In PRISM, the input series is itself obtained from the image through an image-to-sequence map. Consequently, the HVG corresponds to the composition

$$v_{\text{HVG}} = \sigma^{(2)}_{\epsilon_2}\left(\sigma^{(1)}_{\epsilon_1}(v_{\text{img}})\right), \tag{A.5}$$

providing a graph representation of the ordering and relative magnitudes induced by the preceding sequential encoding.

Line graphs [9] provide a second form of composed graph representation. Given a graph $G = (V, E)$, its line graph $L(G)$ associates each edge in G with a node in $L(G)$, with two nodes being adjacent whenever their corresponding edges in G share an endpoint. In PRISM, line graphs are constructed from the region-adjacency graphs obtained after image segmentation. Thus, the transformation follows the composition

$$v_{\text{line}} = \sigma^{(2)}_{\epsilon_2}\left(\sigma^{(1)}_{\epsilon_1}(v_{\text{img}})\right), \tag{A.6}$$

where the first map encodes spatial regions and their adjacency, while the second re-expresses the relations between those adjacencies.

Appendix B. PRISM features

Table B.1: Summary of the feature families extracted from each representation. Region and line graphs are each extracted twice, once per segmenter (SLIC and Felzenszwalb), yielding 2×71 and 2×63 features, respectively. The eight visual representations total 639 features.

Representation	Feature family	No.	Description
Image descriptors (130)	Color	90	Lab histograms, 30 bins per channel.
	Statistics	9	Mean, standard deviation and skewness per Lab channel.
	LBP	18	Uniform Local Binary Patterns ($R = 2$, $P = 16$).
	Haralick	13	Gray-level co-occurrence matrix descriptors.
Region graph (71 per segmenter)	Region statistics	18	Area distribution (7), area-weighted Lab colour and chroma (9), spatial density (2).
	RAG structure	14	Order, size and density (3); degree distribution and hubs (6); edge colour and spatial distances (5).
	k-NN graph	13	Same structural and edge-attribute measures on the k-nearest-neighbour graph ($k = 10$).

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Table B.1 – continued from previous page

Representation	Feature family	No.	Description
	Delaunay / MST	26	Same 13 measures on the Delaunay triangulation and on the minimum spanning tree.
Line graph (63 per segmenter)	Topology	11	Order, size, density, degree distribution, degree assortativity, wedge load concentration.
	Boundary attributes	22	Quantiles of boundary colour contrast and spatial extent, endpoint area ratio, shared area, dispersion and orientation coherence.
	Junction geometry	30	Junction angles, acute/straight/reversal fractions, contrast and spatial deltas between adjacent boundaries, outer-region distances.
Space-filling curve (57)	Sub-band statistics	48	Mean and standard deviation of 8 sub-bands of the Hilbert-ordered 1D signal, per Lab channel.
	Global statistics	6	Mean and standard deviation of the full signal, per Lab channel.
	Channel coupling	3	Pairwise correlation between the three Lab signals.
Spectral (152)	Log-polar power	144	Log power spectrum on a 12 × 12 radius × orientation grid.

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Table B.1 – continued from previous page

Representation	Feature family	No.	Description
	Spectral summaries	8	Radial centroid and spread, low/high frequency fractions, spectral entropy and slope, orientation entropy and peak fraction.
Gabor (32)	Filter bank	32	Mean and standard deviation of the response magnitude for 4 orientations $\times$ 4 frequencies (Manjunath & Ma, 1996).
Text (Variable)	TF-IDF SVM scores	Variable	Label-aligned LinearSVC decision-function scores from a binary TF-IDF (unigram/bigram) classifier trained on generated image captions, one score per style class.

Appendix C. Additional Robustness Analyses

Appendix C.1. Encoding-Parameter Robustness

For a fixed transformation family f, we expect the behavior of $\sigma_{f,\theta}$ to remain stable across a range of admissible implementations $\theta \in \Theta_f$. Different choices of θ may induce trade-offs between computational cost and predictive performance, but the usefulness of a transformation should not depend on a single narrowly tuned configuration θ^*.

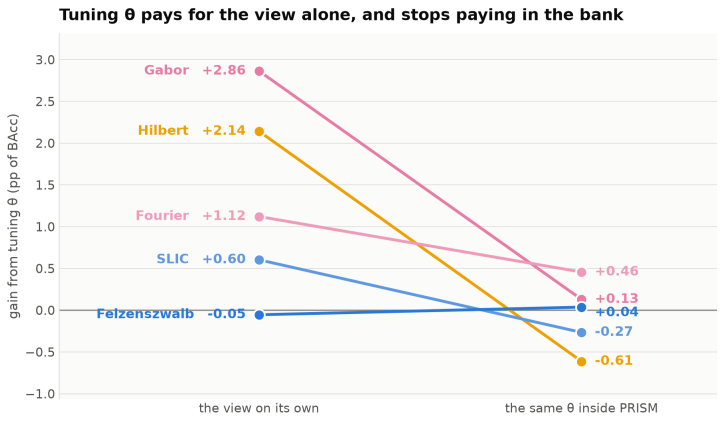


Figure C.1: Effect of encoding-parameter tuning on standalone views and on the same views when incorporated into PRISM. While tuning can considerably improve an isolated representation, its marginal effect becomes small once the view is combined with the remaining PRISM representations.

Figure C.1 evaluates this sensitivity by comparing the gain obtained from tuning θ when each view is used independently and when the same view is integrated into PRISM. Parameter selection can substantially affect the standalone performance of individual views, with gains of up to 2.86 percentage points in balanced accuracy. However, these gains largely vanish once the views are combined within PRISM: the corresponding changes range from small positive improvements to minor degradations. This suggests that the multi-view representation is considerably less sensitive to the precise parameterization of any individual encoding map than the isolated views themselves. In other words, tuning remains important for maximizing the performance of a single representation, but becomes substantially less critical when multiple complementary views are jointly available.

Appendix C.2. Text Generation Setup and leakage analyses

To obtain an affect-oriented text signal that is provably free of style or artist information, we distill a compact caption generator through a teacher-student protocol entirely decoupled from WikiArt. A vision-language teacher (Qwen2-VL-7B-Instruct) is prompted, at temperature 0, to describe each COCO train2017 image in two to three sentences focused strictly on the emotional tone conveyed by visible color, light, composition and expression, explicitly instructed not to invent narrative content or name any style, artist, or place; the resulting captions, truncated to their last complete sentence and terminated with an explicit end-of-sequence token, form the distillation targets.

A byte-pair tokenizer (vocabulary size 12,000) is fit on this teacher corpus with four reserved special tokens (`<pad>`, `<bos>`, `<eos>`, `<img>`). The student is a compact decoder-only Transformer trained from scratch (8 layers, 8 heads, d_{model} = 512, feed-forward dimension 2,048, dropout 0.1) conditioned on a frozen CLIP ViT-B/32 image embedding projected into a 10-token visual prefix, following the ClipCap paradigm; it is optimized for 15 epochs (batch size 256, learning rate 2×10^{-4}, weight decay 0.01, label smoothing 0.1) with early stopping on validation ROUGE-L.

Because the student never observes a WikiArt image, an artist name, or a style label during training, its captions constitute a clean affective control when subsequently applied to WikiArt: generation fidelity is first verified against the teacher on a held-out COCO test split (BLEU-4, ROUGE-L), after which the same decoder is run on WikiArt images to produce captions that are audited for residual style/artist leakage (literal mention of the ground-truth style token, of any of the 27 WikiArt style names, or of the artist's surname) and evaluated for downstream style separability using the identical stop-word-scrubbed, binary TF-IDF (n-grams 1–2, min_df = 10, max_df = 0.90, 50,000 features) plus L_1-regularized linear SVM (C = 0.1, class-balanced, 5-fold stratified out-of-fold) protocol used elsewhere in the paper, so that its balanced accuracy can be directly compared to the project's CATR/COCO caption baseline as a lower-bound reference for text-derived style signal.

Table C.2 shows that CLIP and Qwen2-VL are, on their own, surprisingly good at guessing Multitask style. With no training and a single generic prompt per class, CLIP reaches BAcc = 0.325 with ViT-B/32 and 0.394 with ViT-L/14, which corresponds to roughly 9 to 11 times chance. Qwen2-VL, asked blindly to simply name a painting style, reaches approximately 8 times chance even in a small pilot cohort. Neither model has ever been trained on this task.

The most plausible explanation is memorization rather than genuine stylistic reasoning. Both

Table C.2: Style-leakage probes on Multitask125 style prediction.

Condition	**BAcc**	**Acc.**	**Macro-F1**
CLIP zero-shot (ViT-B/32, *openai*)	11.9 ± 0.6	8.6 ± 0.1	5.9 ± 0.3
CLIP zero-shot (ViT-L/14, *openai*)	16.6 ± 0.2	10.9 ± 0.2	8.6 ± 0.2
Qwen2-VL-7B blind style naming[†]	**28.4 ± 0.8**	**28.5 ± 0.4**	**21.3 ± 0.5**
CATR/COCO caption + TF-IDF/SVM (clean control)	9.0 ± 0.4	5.0 ± 0.2	3.8 ± 0.2
Distilled student caption (COCO-only) + TF-IDF/SVM[‡]	16.1 ± 0.6	14.8 ± 0.4	9.6 ± 0.3

[†] Pilot cohort, style-stratified, free-text answer scored by literal alias match against the ground-truth label, reported as recall, not a trained classifier's BAcc, and therefore not directly comparable in scale to the other rows. [‡] Same 15,860-image Multitask test split as the CLIP rows; the true style name appears in only 1.7% of captions (vs. near-zero for CATR/COCO), consistent with a small residual leak.

models were pre-trained on massive web corpora that likely contain Multitask images paired with the exact style and artist names used in this benchmark. The Qwen probe supports this interpretation because, besides recovering the painting style, it retrieves artist surnames and near-verbatim artwork titles directly from pixels, which indicates recall of previously seen examples rather than inference from visual style alone.

For this reason, neither model is used directly as a style or affect feature, since doing so risks introducing target label leakage into the downstream pipeline. Instead, CLIP is used only as a frozen image encoder, while Qwen2-VL serves solely as a teacher whose prompt explicitly forbids mentioning style, artist, or location. The generated captions are subsequently distilled into a student model and evaluated only through out-of-fold predictions. Under this protocol, the distilled student reaches BAcc = 0.242 on the same Multitask test split, which is substantially closer to the clean CATR/COCO baseline of 0.181 than to the off-the-shelf CLIP models ranging from 0.325 to 0.394. Moreover, the true style name appears in only 1.7% of the student captions, which indicates that although a small residual leak remains, it is substantially smaller than the direct leakage observed in CLIP or Qwen2-VL.

Table C.3 provides an audit of text leakage across captioners. In this analysis, movement terms are decomposed because `abstract` is a common English adjective, so describing an abstract canvas as abstract is better interpreted as descriptive visual perception rather than explicit label leakage, and removing it reduces channel performance by only 0.13 percentage points. The term `abstract` is the most generic class that can be associated with almost any data source,

Table C.3: Frequency of art-movement and artist-name occurrences across captioners over 99,816 images. Counts indicate captions containing at least one matching token after standard text normalization.

	Movement name		
Captioner	**Any term**	`abstract` **only**	**Excluding** `abstract`
CATR/COCO	6 (0.01%)	5 (0.01%)	1 (0.00%)
Distilled student	5,929 (5.94%)	5,727 (5.74%)	202 (0.20%)
Qwen2-VL-7B	15,433 (15.46%)	13,916 (13.94%)	1,517 (1.52%)

	Artist name	
Captioner	**Strict**	**Surname**
CATR/COCO	0 (0.00%)	2 (0.00%)
Distilled student	0 (0.00%)	6 (0.01%)
Qwen2-VL-7B	63 (0.06%)	109 (0.11%)

which makes it particularly prone to false accusations of cross-data leakage. For instance, consider a classifier that only uses labels such as bird, dog, and others, where any animal, or even a target object under poor image conditions, might be assigned to the others class. If two independent classifiers maintain this consistency, it results in a false positive accusation of data leakage. This concept of handling generic or unseen classes is widely studied in open-world classification literature [5], and we believe this scenario falls into a similar gray area. Furthermore, artist matching is divided into strict matching, which requires the last two tokens of the artist name to appear contiguously like van gogh, and surname matching, which requires only the final token. Surname matching serves as an upper bound because common English words such as Bacon, Wood, and Church frequently appear as standard vocabulary. Painting titles cannot be audited in this corpus because the ground truth provides numeric identifiers alongside filename, artist, style, and genre. As a final safety check, the textual channel is included as a supplementary input within the pipeline rather than a primary dependency, which ensures that the overall architecture does not depend strictly on text representations.

Appendix C.3. Pretraining parameters

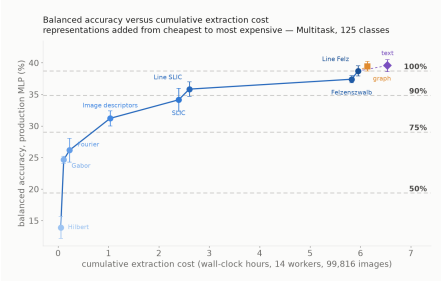
Fine-tuning configuration for both backbones. All image operations use bicubic interpolation and ImageNet channel statistics. Class weights are $w_c \propto 1/n_c$, normalised to unit mean; this targets balanced accuracy and deliberately trades simple accuracy for it, which is why all three metrics are reported. *Early-stop patience was initially 12 (ResNet-50) and 10 (ViT-B/16); for the two corpora trained last (Places-LT, Rijksmuseum) it was set equal to the epoch budget, because a short patience interacts badly with cosine annealing — the mid-run plateau is a property of the schedule rather than evidence of convergence. Food-101, MINC-2500, Multitask-125, NCT-CRC and OilPainting retain the shorter setting and were not re-run. Every other hyperparameter is identical across the fourteen corpus × backbone cells. A shorter patience can only weaken the fine-tuned baselines, so where it remains it works in favour of the fusion results we report; and in Places-LT, where the full budget was available, the selected epoch was still 8 of 75 (ResNet-50) and 6 of 40 (ViT-B/16), indicating that early stopping was not the binding constraint there.

Table C.4

Setting	**ResNet-50**	**ViT-B/16**
Shared		
Initialisation	ImageNet-1k, all layers trainable	
Resolution	224×224	
Train augmentation	RandomResizedCrop(0.6–1.0) + flip + RandAugment$(2, 9)$	
Eval transform	Resize(256) + CenterCrop(224)	
Loss	cross-entropy, class weights $\propto 1/n_c$	
Scheduler	cosine, $T_{\max}$ = budget, no warm-up	
Precision	mixed (AMP)	
Model selection	best epoch by validation balanced accuracy	
Validation	official `val`, else stratified 10% of train	
Repetitions	5 seeds; 5 folds if no published split	
Per architecture		
Optimiser	SGD (Nesterov)	AdamW
Learning rate	0.01	1×10^{-4}
Weight decay	1×10^{-4}	0.05
Momentum	0.9	—
Label smoothing	—	0.1
Batch size	128	64
Epoch budget	75	40
Early-stop patience	12 / 75*	10 / 40*

Appendix C.4. Time complexity analisys

Figure C.2a shows the cumulative extraction cost and predictive performance as views are progressively added. Two regimes can be observed. In the early portion of the curve, additional views incur relatively low computational cost while producing substantial gains in balanced accuracy. As more views are incorporated, however, the curve gradually saturates, and the marginal performance improvement decreases, indicating diminishing returns.



(a) Cumulative cost and performance of the views ensemble.



(b) Asymptotic analyses of each view.

Figure C.2: Asymptotic and measured time of PRISM.

Two aspects should be highlighted when interpreting this behavior. First, do not attribute the steep initial growth specifically to the Hilbert and Gabor representations. Rather, it reflects the general diminishing-return behavior expected when informative views are progressively added, and a similar pattern would emerge under alternative orderings. Second, the final two views have distinct computational characteristics. The textual representation relies primarily on GPU-based inference, whereas the global similarity graph can only be constructed after the remaining feature representations have been extracted, since its definition depends on them.

Figure C.2b shows that the computational cost of the different views scales differently with input resolution. While some transformations, such as Felzenszwalb, Gabor, and Hilbert, grow approximately quadratically with image side length, others, including SLIC and the image-descriptor branch, scale more mildly. This indicates that the relative computational attractiveness of each representation is resolution-dependent rather than fixed. A final remark about the cost is that we do not optimize the views; both SLIC and Felzenszwalb have substitute methods that remain competitive with better cost.

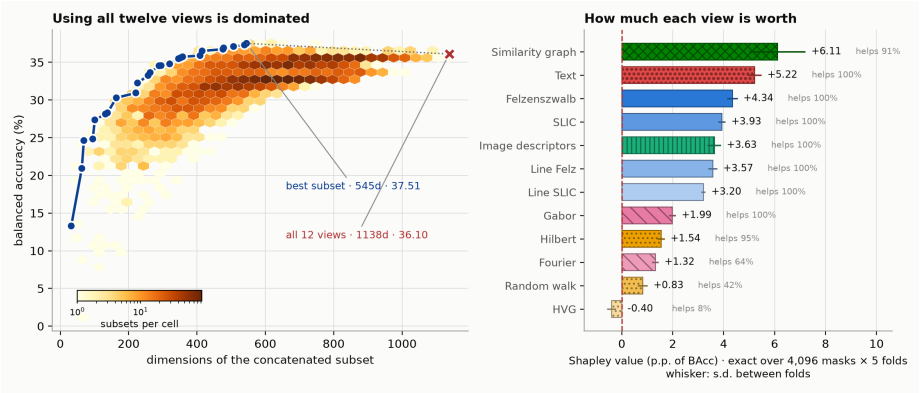


Figure C.3: Pareto frontier and Shapley values for the multitask125 dataset.

Appendix C.5. Shapley analyses

Figure **??** shows the Pareto frontier obtained by enumerating all 2^{12} view subsets under the LDA probe. Not every view is equally useful, and the right panel separates two failure modes that a single marginal value conflates. HVG is consistently harmful: adding it degrades the set in 92% of all 2048 coalitions. Random walk and Fourier are instead inconsistent, helping in 42% and 64% of coalitions, respectively, against more than 95% for every other view. All three are absent from every point of the frontier.

The frontier also shows that using all twelve views is dominated: the best subset reaches 37.51 with 545 dimensions, against 36.10 with all twelve and 1,138 dimensions (+1.41 pp paired by fold, $p = 0.011$, 5/5 folds). We do not, however, read the exact composition of that maximum as a recommendation. Near the top of the frontier, adding or removing a single view moves balanced accuracy by 0.1–0.3 pp, which is well inside the between-fold standard deviation of $\approx$ 1.1 pp; removing Fourier alone from the full set, for instance, is not significant (+0.12 pp, $p = 0.25$). The enumeration identifies which views never earn their place, not which of several near-optimal subsets is best.

Two properties of the probe should be kept in mind. LDA projects onto at most $C - 1$ discriminant directions, so subsets differing in dimensionality are not compared on equal footing at the top of the range, and the estimate of the pooled covariance degrades as the concatenated dimension grows. The enumeration is therefore used to rank subsets and to identify views that never contribute, not to predict the accuracy a higher-capacity head would reach on any particular subset.

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